MIL-B-43021D
31 May 1978
SUPERSEDING
MIL-B-43021C
5 March 1968

MILITARY SPECIFICATION

BINDER, LOAD, CARGO TIE-DOWN, 10,000 POUNDS (44.5 kN) CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size load binder of 10,000 pounds (44.5 kN) capacity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets

DRAWING

US ARMY RESEARCH AND DEVELOPMENT COMMAND

11-1-901 - Binder, Load, Cargo Tie-Down, 10,000 Pounds (44.5 kN) Capacity, Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.2 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial load binders provided the complete item complies with all provisions of this specification (see 6.4).

3.3 Design and construction. Materials, components, protective finish, marking, construction and assembly shall conform to Drawing 11-1-901.

3.4 Performance.

3.4.1 Proof load. The assembled load binder shall show no visible evidence of release, fracture, weld separation, or permanent deformation when proof load tested to 10,000 pounds (44.5 kN) as specified in 4.5.1.

3.4.2 Ultimate load. The assembled load binder shall withstand an ultimate load of 13,000 pounds (57.8 kN) without failure when tested as specified in 4.5.2. Deformation but not rupture of parts shall be allowed.

3.5 Identification marking. Each load binder shall be marked for identification with the capacity, 10,000 LB, manufacturer's name or readily identifiable trade mark, and the Part Number 11-1-901. Part numbering shall be in accordance with MIL-STD-961. The markings shall be permanently indented, located as shown on the drawing, and of such size as to be clearly legible.

3.6 Workmanship. All components and assemblies of the load binder assembly shall be free from dirt and other extraneous material, burrs, slivers, dents, cracks, fractures and damaged welds. External surfaces shall be free from sharp edges and corners except when required. Components shall not be buckled, loose, dented, missing, malformed or misaligned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 6.2), the load binder shall be examined for defects in table I and tested as specified in 4.5.1 and 4.5.2. The presence of any defect or the failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 End item inspection. The lot shall consist of all load binders offered for inspection at one time. The sample unit shall be one load binder.

4.4.2.1 Visual examination. An examination shall be made of the load binders for defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish, protective	Any breaks through coating of metal		X
	Any area of rust or corrosion	X	
	Dirt, oil, grease, or other foreign matter		X
Construction and workmanship	Any component missing	X	
	Any component cracked, fractured or otherwise damaged, including damaged welds	X	
	Any sharp edges, burrs or metal slivers	X	
	Any part not assembled or joined as specified	X	
Identification marking	Missing, incomplete not legible, incorrect or not located or applied as specified		X

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4.4.2.2 Dimensional examination. An examination shall be made of the load binders to determine compliance with dimensions specified. Any dimensions not meeting specified requirements shall be a defect. The inspection level shall be S-3 with an AQL of 2.5 expressed in terms of defects per hundred units.

4.4.2.3 End item testing. Testing shall be performed as specified in 4.5.1 and 4.5.2 for conformance to the requirements specified in 3.4.1 and 3.4.2 respectively. The sample unit shall be a complete load binder assembly. The inspection level for the proof load test shall be S-2 with results reported as pass or fail and an acceptance number 0. The inspection level for the ultimate load shall be two per thousand with results reported as pass or fail and an acceptance number 0. Failure of a test sample shall be cause for rejection of the lot. Acceptance of quantities on hand or subsequently produced shall be withheld until cause of failure is determined and appropriate corrective action taken to prevent recurrence.

4.4.3 Inspection of packaging. An examination shall be made to determine that preservation-packaging, packing and marking as required by section 5 are complied with. Defects shall be as indicated in table II. The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one shipping container fully prepared for delivery except it need not be closed. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of packaging

Examine	Defect
Markings (exterior) and interior container)	Omitted, incorrect, not legible, of improper size, location, sequence or method of application
Materials	Component missing, damaged, or defective
Workmanship	Contents not compact in containers, incomplete closure of container flaps Bulging or distortion of container
Contents (exterior and interior container)	Number per container not as specified

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4.4.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.4.4 Examination for palletization. An examination shall be made to determine that palletization is in compliance with 5.3. Defects shall be scored as specified in table III. The sample unit for this examination shall be one palletized unit load, ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE III. Examination of palletization

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application

4.5 Tests.

4.5.1 Proof load test. The binder shall be secured in a suitable tensile testing device and subjected to a static load of 10,000 pounds (44.5 kN) applied for not less than 30 seconds. Direction of load shall be as shown on Drawing 11-1-901. After load relaxation, the hooks shall be disassembled from the binder and all components examined for evidence of fracture, weld separation and permanent deformation. Any change in excess of 1/32 inch (0.79 mm) from the fabricated dimensions, in those dimensions parallel to the direction of stress, shall be classified as visible permanent deformation.

4.5.2 Ultimate load test. The binder shall be secured in a suitable tensile testing device and subjected to a static load of 13,000 pounds (57.8 kN) applied for not less than 30 seconds in the direction shown on Drawing 11-1-901. After load relaxation, the binder shall be examined for evidence of damage cited in 3.4.2.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each binder shall be individually preserved-packaged in a fiberboard box conforming to style OPF, grade W6c of PPP-B-636. Inside dimensions of each box shall approximate 13-1/4 inches in length, 3-5/8 inches in width, and 1-1/8 inches in depth (337 mm in length, 92 mm in width, and 29 mm in depth). Approximate dimensions are furnished as a guide only. Each interior box shall be closed method IV and reinforced method III in accordance with the appendix of PPP-B-636.

5.1.2 Commercial. Each binder shall be preserved-packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A packing. Seventy-two binders, packaged as specified in 5.1.1 shall be packed in a shipping container conforming to overseas type, style A or J, type 2 load of PPP-B-601; or class 3, style optional, type 2 load of PPP-B-585. The packages shall be packed on end, 6 in length, 12 in width, and 1 in depth (152 mm in length, 305 in width and 25 mm in depth) within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/4 inches in length, 16-1/2 inches in width, and 13-1/2 inches in depth (591 mm in length, 305 mm in width and 343 mm in depth) within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/4 inches in length, 16-1/2 inches in width, and 13-1/2 inches in depth (152 mm in length, 305 mm in width and 25 mm in depth). Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

5.2.2 Level B packing. Seventy-two binders, packaged as specified in 5.1.1, shall be packed in a shipping container conforming to domestic type, style A or J, type 2 load of PPP-B-601; or class 1, style optional, type 2 load of PPP-B-585. The packages shall be packed on end, 6 in length, 12 in width, and 1 in depth (152 mm in length, 305 mm in width and 25 mm in depth) within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/4 inches in length, 16-1/2 inches in width, and 13-1/2 inches in depth (591 mm in length, 305 mm in width and 343 mm in depth). Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

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5.2.3 Commercial packing. Binders, preserved-packaged as specified in 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution.

5.3 Palletization. When specified (see 6.2), binders, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The load binder covered by this specification is designed for use as a tiedown tensioning device in rigging general cargo and equipment for transport and air delivery.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) When a first article is required (see 3.1, 4.3 and 6.3)
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
(d) When palletization is required (see 5.3)

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one complete load binder assembly. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US Customary Units shall be controlling.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - AV
Air Force - 11

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 1670-0463

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-B-43021D
Binder, Load, Cargo Tie-Down, 10,000 Pounds (44.5 kN) Capacity

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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